Tunable 433.92 MHz ring oscillator: GF180 candidate options

Matt Venn / Zero to ASIC Course 2026-11-07 Preliminary

Pre-silicon · schematic simulation (ngspice, gf180mcuD nfet/pfet_03v3) · not extracted

Description

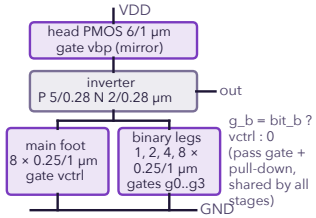
Option A uses about 2.2× less power than **Option B** and rejects supply changes about 2.7× better; B gives the complementary outputs the antenna drive needs. As drawn, both tune far too steeply with vctrl; a constant-gm bias fixes this. A plain inverter ring is the untuned baseline.

Both are current-starved ring VCOs (voltage-controlled oscillators) for a Tiny Tapeout GF180MCU transmitter. Each inverter's current is set by a PMOS head and an NMOS foot of binary legs switched between **vctrl** and 0: the 4-bit **code** tunes coarsely, vctrl finely. A is a 5-stage single-ended (SE) ring; B is two 4-stage rings tied by weak cross-coupled inverters.

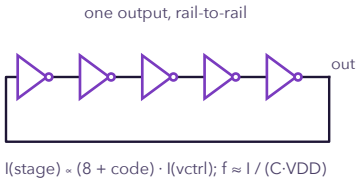
Features

- Coarse: 4-bit binary foot legs (k = 8 base), ×2.06 (A) / ×2.89 (B) at vctrl 1.5 V
- Fine: vctrl on every leg; 117–1,039 MHz (A), 146–832 MHz (B)
- Power at 433.92 MHz: A 0.91 mW, B 2.03 mW (plain ring 2.46 mW)
- A + constant-gm bias (a β-multiplier, whose current is set by an on-chip resistor, not vctrl): 5-bit coarse, ±3.8 % fine; Kvco (frequency change per volt of vctrl) about 16 MHz/V vs 782 for A
- 3.3 V devices only, no inductors, small area

Starved stage (both options)

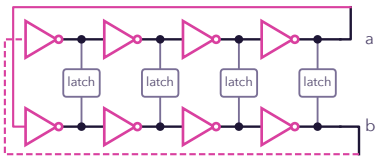


Option A: 5-stage single-ended ring



I(stage) ∝ (8 + code) · I(vctrl); f ≈ I / (C·VDD)

Option B: 4-stage pseudo-differential

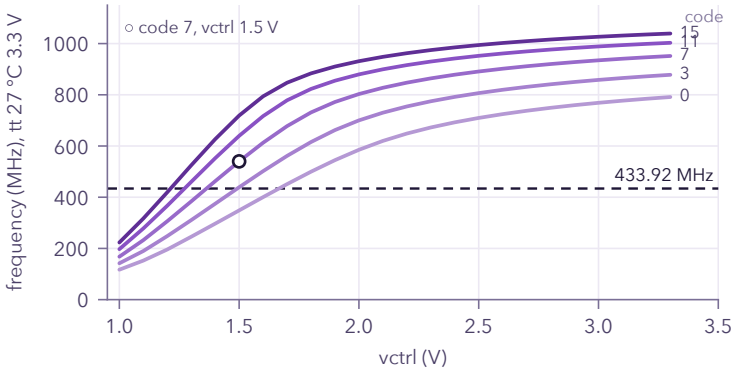


wrap crosses a↔b; latch = weak back-to-back inverters (wx 0.5 µm)

Key specifications

tt, 27 °C, 3.3 V unless noted	A · SE N=5	B · diff N=4	A + const-gm bias	Plain ring (baseline)
Tuning span (all codes, vctrl)	117–1,039 MHz	146–832 MHz	**236–728 MHz**	none
Coarse range, top / bottom code	×2.06 (1.5 V)	×2.89 (1.5 V)	**×2.84, about 15 MHz steps**	–
Fine tuning: Kvco at 433.92 MHz	782 MHz/V	522 MHz/V	**about 16 MHz/V, ±3.8 %**	–
df/dVDD, %/10 mV (kHz at 433.92 MHz)	−0.11 (−474)	−0.29 (−1,269)	**−0.27 (−1,170)**	+0.23 (+993)
Tempco, 0–70 °C	−2,197 ppm/°C	−1,982 ppm/°C	**+2,952 ppm/°C**	−1,843 ppm/°C
PVT spread (process, VDD, temp.)	×0.68 … ×1.42	×0.62 … ×1.52	**×0.78 … ×1.23**	×0.73 … ×1.29
Power at 433.92 MHz	0.91 mW	2.03 mW	**1.59 mW at 508 MHz**	2.46 mW
Outputs	1, single-ended	2, complementary	**1, single-ended**	1, single-ended

Option A: single-ended, N=5



Option B: pseudo-differential, N=4

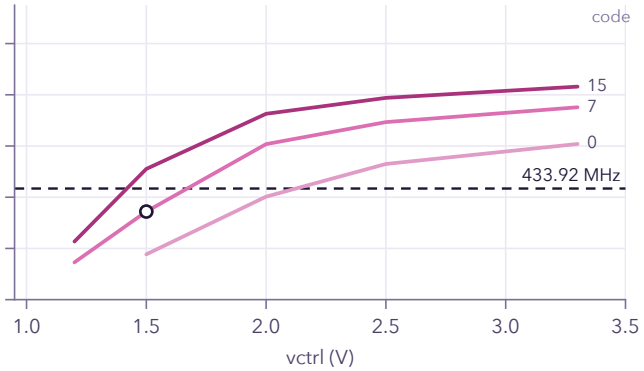


Fig. 1. Frequency vs vctrl for codes 0, 3, 7, 11, 15 (light → dark), tt 27 °C 3.3 V; at code 7, A reaches 433.92 MHz at 1.36 V and B at 1.67 V. Points below about 110 MHz are outside the measurement window (10 periods in 100–200 ns) and are not plotted. B: coarse sweep (codes 0/7/15, vctrl 0.9/1.2/1.5/2.0/2.5/3.3 V), so its curves and Kvco are coarse.

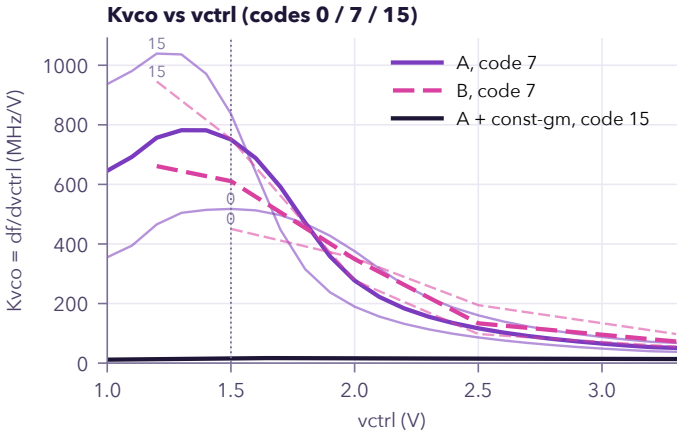


Fig. 2. Kvco vs vctrl: about 782 MHz/V for A at 433.92 MHz, about 16 MHz/V with constant-gm bias (black). A solid, B dashed; code 7 bold, codes 0 and 15 faint.

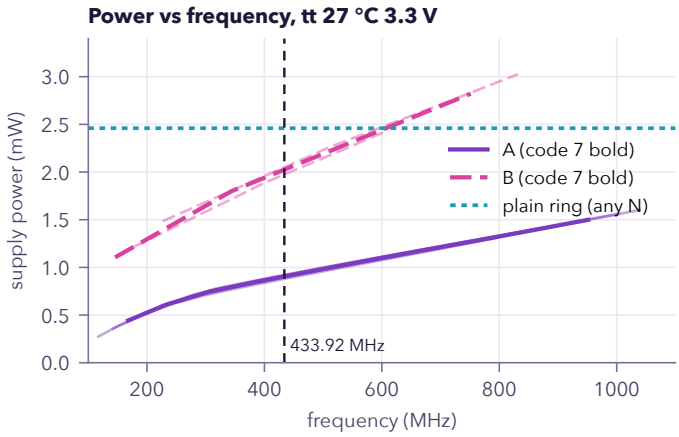


Fig. 3. Supply power vs frequency for every code: A draws 0.91 mW at 433.92 MHz, B 2.03 mW. The plain ring draws about 2.5 mW whatever N is.

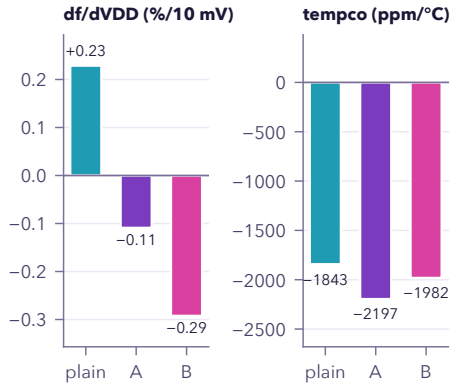


Fig. 4. Supply sensitivity (±150 mV) and tempco (0–70 °C, fixed vctrl) at tt; A is lowest at −0.11 %/10 mV.

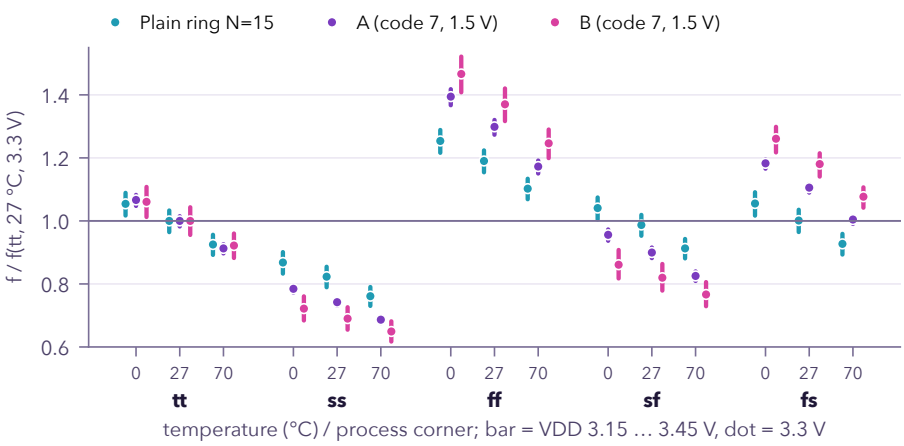


Fig. 5. PVT (process, voltage, temperature) spread, each normalised to its own tt 27 °C 3.3 V; A spans ×0.68 to ×1.42. A and B at code 7, vctrl 1.5 V.

Trade-offs

Option A: single-ended, N=5

- Lower power: 0.91 mW at 433.92 MHz vs 2.03 mW for B (2.09 vs 4.67 µW/MHz).
- Better supply rejection (PSRR): −0.11 vs −0.29 %/10 mV, opposite in sign to the plain ring as the current is fixed and the swing is VDD.
- Kvco at 433.92 MHz (code 7): 782 MHz/V vs 522 for B (higher: A sits lower on its curve, at 1.36 V).
- One output only: the antiphase antenna drive needs an extra inverter, which adds skew.

Option B: pseudo-differential, N=4

- Complementary outputs a0 / b0 for the antiphase antenna drive, no extra splitter.
- Possibly less flicker upconversion (low-frequency transistor noise turned into phase noise), as symmetric waveforms give a smaller ISF (impulse sensitivity function) average; **needs a phase-noise simulation**.
- No tail current source, so supply rejection is about 2.7× worse than A.
- About 2.2× the power of A at 433.92 MHz and 1.7× the devices.

Sizes: core Wn/Wp 2/5 µm, L 0.28 µm; starving L 1 µm. Plain-ring figures from N=15. Tuning span: codes 0–15, vctrl 1.0–3.3 V (A + const-gm: codes 0–31, vctrl 0–3.3 V; PVT and sensitivities at code 15, vctrl 1.65 V). Data: `sim/vco/`. Regenerate: `python3 docs/datasheet/make_datasheet.py`.